HALO

Holographic Artificial-Life with Observational Confinement — Plain-Language Summary

Quantum Computing | Artificial Life | Grover Amplitude Amplification | Holographic Compression | Six-Rung Evidence Ladder

1. What is HALO?

HALO is a blueprint for building **artificial life entirely inside a quantum computer**. It combines four well-understood quantum techniques into a single architecture: a clock that runs without any external timekeeper, noise-protected agent states, memory-efficient holographic compression, and a fitness test that is computationally hard for outside observers to reverse-engineer.

The core idea is that all the machinery of evolution — time, noise protection, state compression, and fitness selection — can be implemented as sequences of quantum gates, with no classical computer stepping in between.

This document explains what has actually been demonstrated, step by step, and what remains to be done. Every claim is evaluated on a six-rung evidence ladder.

2. The Six-Rung Evidence Ladder

Rather than making a single large claim, HALO uses a **claim ladder**: six levels of assertions, each strictly stronger than the last. Each rung carries a declared evidence threshold — a concrete test that can falsify it. Rungs C0 through C5 have been validated; C6 is explicitly deferred to future work.

C0 **Toy composability** VALIDATED

A 5-qubit noiseless circuit confirms that the clock, DFS noise protection, holographic map, and fitness verifier all work together. Ancilla qubits return cleanly to zero (no irreversible erasure).

C1 **Scalability proxy** VALIDATED

Binary-encoded circuits scale sub-linearly in qubit count (log-log slope 0.43 vs. 1.0 for one-per-site baseline) on a lattice-Trotter benchmark. A depth tradeoff (~2.9x deeper circuits) is openly reported.

C2 **Subspace compression** VALIDATED

Explicit isometric embeddings for k-excitation subspaces (k = 1, 2, 3; bulk sizes up to N = 32) achieve zero isometry error across all 12 instances, with compression ratios up to 6.4x.

C3 **Security hardening trend** VALIDATED

A bounded heuristic attacker's success rate drops from 69% to 20% as system size grows from n = 6 to n = 18 qubits (Mann-Kendall p = 0.0027). This is an empirical trend, not a complexity-theoretic proof.

C4 **Cross-sector holographic compression** VALIDATED

Direct-sum isometries spanning multiple excitation sectors simultaneously (K in {1,2}, {0,1,2}, {1,2,3}) are constructed for bulk sizes N in {4, 6}. All 6 instances achieve zero isometry error. Sector leakage under a transverse-field perturbation is characterised: exactly zero at strength = 0; for short evolution times (t=0.5) leakage grows monotonically with perturbation strength; at longer times (t=1.0, t=2.0) leakage peaks near $\Gamma \approx 0.625$ then decreases, consistent with quantum recurrence effects.

C5 **Fully quantum multi-generation evolution** VALIDATED

A Qiskit 5-qubit circuit evolves P = 8 agents through G = 5 generations with no intermediate measurement. Fit-agent

probability oscillates between 0.25 and 0.82 — the signature of Grover amplitude amplification. The full selection-plus-mutation loop runs coherently from start to finish.

C6 **Hardware validation and universal compression** DEFERRED

Universal holographic compression for arbitrary Hamiltonians, unconditional cryptographic security, and physical hardware experiments at scale. These require advances beyond the current scope.

3. Key Results at a Glance

6
Claim rungs validated (C0-C5); C6 explicitly deferred

6.4x
Best compression ratio achieved (C2, N=32, k=1), zero error

0.43
Qubit-count log-log slope for binary encoding vs. 1.0 for one-hot (C1)

p=0.003
Mann-Kendall p-value for hardening trend under heuristic attack (C3)

6
Cross-sector isometry instances, all zero error (C4)

0.82
Peak fit-agent probability via Grover evolution (C5), up from 0.25

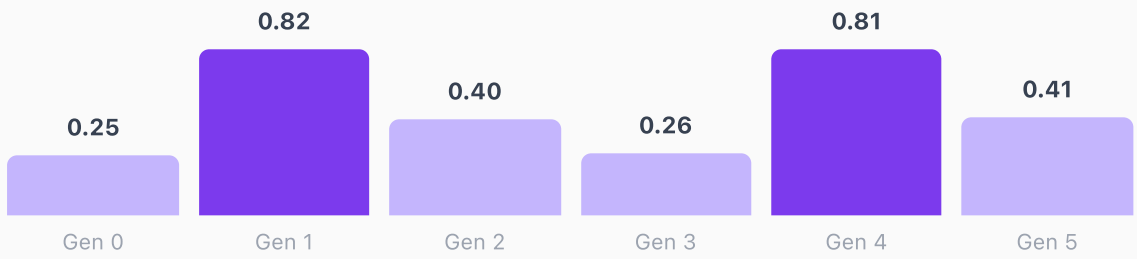
4. How Does the Quantum Evolution Work? (C5)

Imagine 8 agents placed in a quantum superposition — all of them existing in the circuit simultaneously. Each agent has an angle encoding its "fitness". Two agents (numbers 6 and 7) start near the ideal angle; the other six are farther away.

One generation of evolution applies three quantum gates in sequence:

1. **Fitness oracle** — a multi-controlled phase flip that marks the two fit agents with a -1 phase, without measuring or collapsing the state.
2. **Grover diffusion** — inversion-about-mean on the population register, which constructively amplifies the fit agents and destructively cancels the unfit ones.
3. **Coherent mutation** — a random DFS rotation applied independently to each agent's angle, keeping the state fully coherent throughout.

The result is a characteristic oscillation: the probability of landing on a fit agent shoots up on odd generations and partially falls back on even ones — exactly the Grover over-rotation cycle. The chart below shows this pattern.



Fit-agent probability P_fit over 5 generations (C5 result). The initial fit fraction is 0.25 (2 out of 8 agents). Odd generations amplify fit agents; even generations show Grover over-rotation. No intermediate measurement is performed throughout.

5. What Does Cross-Sector Compression Add? (C4)

The C2 result showed that quantum states in a *single* excitation sector (e.g., exactly 1, or exactly 2, or exactly 3 particles present at a time) can be compressed into far fewer qubits with no information loss.

C4 extends this to states that are *quantum superpositions across multiple sectors at once* (e.g., both 1-particle and 2-particle components simultaneously). A *direct-sum isometry* handles the combined subspace. For example, combining sectors {1,2,3} at N = 6 sites compresses a 41-dimensional subspace into just 6 qubits (2^6 = 64), with zero isometry error.

The leakage experiment answers a practical question: how quickly does a state *leave* its sector when a perturbation is switched on? The answer is clean: at zero perturbation strength, leakage is exactly zero for any evolution time. At short evolution times it grows monotonically with perturbation strength. At longer times

it peaks near $\Gamma \approx 0.625$ then decreases — a quantum recurrence effect. Together these results give a precise quantitative boundary for when the single-sector approximation breaks down.

6. What This Paper Does Not Claim

Explicitly deferred to future work (C6 and beyond)

- ✗ Universal holographic compression for arbitrary Hamiltonians without sector restriction
- ✗ Cryptographic security proof against polynomial-time quantum adversaries
- ✗ Any experiment on physical quantum hardware (all results are noiseless simulation)
- ✗ Noise-resilient composition under realistic NISQ hardware error models
- ✗ Multi-generation heredity, crossover, or multi-locus fitness landscapes
- ✗ Population evolution scaling to P much larger than 8 agents

The depth tradeoff for C1 is also openly acknowledged: the binary encoding cuts qubit count by 57% on average, but increases two-qubit circuit depth by a factor of 2.9. In depth-limited hardware this tradeoff is unfavourable.

7. The Four Quantum Primitives

Every component of HALO rests on an independently-established quantum information result:

Page-Wootters clock. Instead of an external clock, a single qubit is prepared in superposition and entangled with the agent register. "Time" emerges from this entanglement: the clock value conditions whether evolution has happened or not. No classical timer is needed.

Decoherence-free subspace (DFS) encoding. Agent states live in the two-qubit subspace spanned by $|01\rangle$ and $|10\rangle$. Collective environmental noise (same rotation on both qubits) acts as a global phase on this subspace, which is undetectable

under measurement. The agent is passively immune to this noise class without active error correction.

Holographic compression. Inspired by AdS/CFT holography, an isometric map encodes a large "bulk" Hilbert space into a smaller "boundary" register. The C2 and C4 experiments make this concrete: specific subspaces are compressed into log(dim) boundary qubits with zero information loss, verified algebraically.

QMA-hard fitness. The fitness landscape is the ground-state energy of a masked local Hamiltonian. An agent with the right "genetic secret" evaluates its own fitness efficiently; an outside observer faces a problem believed to be hard for any polynomial-time quantum computer. The C3 experiments provide empirical (not complexity-theoretic) evidence for this hardening.

8. Reproducibility

Every number in the paper can be reproduced from the accompanying artifact package. The package contains self-contained Python scripts, JSONL data files, and figure-generation code. C4 and C5 scripts run with a numpy-only fallback when Qiskit is not installed, producing identical numerical results.

halo_simulation.py
C0 toy circuit: three pass/fail tests

scaling_benchmark.py
C1: one-hot vs binary qubit-count scaling benchmark

run_extended_experiments.py
C2-C3: compression & security suites

cross_sector_compression.p
C4: direct-sum isometries and leakage sweep

quantum_evolution.py
C5: Grover evolution loop, 5 generations

data/*.jsonl
Raw results for all six experiments

latex_source/fig_*.png
All figures used in the paper

9. What Comes Next (C6 and Open Problems)

The six-rung ladder is a map, not just a record. The validated rungs define a solid base; the open problems define the next targets.

The most immediate next step is **hardware validation**: running the C0 toy circuit and at least the C5 evolution loop on a physical quantum device (trapped ion or superconducting), to see how noise affects the Grover oscillation and DFS protection in practice.

On the theory side, **average-case hardness** of the fitness function remains unproven. The C3 empirical trend is consistent with hardness, but a reduction from a well-studied problem (such as Learning With Errors) would be needed for a complexity-theoretic guarantee. The paper sketches this conditional reduction as a proposition in Appendix C.

Finally, **multi-epoch population dynamics** — heredity, crossover, multi-locus fitness — require larger circuits than the current 5-qubit toy. The C5 Grover loop is a stepping stone demonstrating that coherent selection and mutation can coexist. Scaling this to populations of 64 or more agents while preserving full coherence is a meaningful near-term target.